

Paul Enrade, pd 7

Game title: Geometry Dash Infinite

Description: This game is meant to resemble the look and feel of Geometry Dash gameplay, except with infinite land generation (like in Google Chrome's dinosaur game). The player must time jumps correctly to dodge spikes and traverse the landscape, and will try to maintain a flawless run for as long as possible. The terrain will become more difficult to traverse as the game progresses, producing more dangerous structures and more frequent obstacles. There will be a score monitor at the top of the screen to keep track of your progress in the game. The game also features the use of The Processing Foundation's Sound library, which is used to play sound effects and soundtracks throughout the game.

All Functionalities:

A. Game mechanics

a. World generation:

- i. LandTile - Each LandTile has a certain elevation and may possess an obstacle and/or a jump pad. If the LandTile is midair, the obstacle may be placed on top or hanging under the LandTile. The jump pad may only be placed on top of the LandTile. If both occupy the same LandTile, each be oriented appropriately.
- ii. JumpOrb - may be placed anywhere in the world (but usually midair), allowing the Player to jump midair given they are within the vicinity of the orb.
- iii. Chunk - A chunk generator produces certain patterns of LandTiles and JumpOrbs to diversify world elements and regulate difficulty of the game. There are three possible difficulties: easy, medium, and difficult.

b. Player-related

- i. 3 kinds of jumps:
 1. NORMAL: base jump power, performed by the player while on the ground.
 2. MINOR: moderate jump power, performed by jumping within the vicinity of a JumpOrb or by jumping midair with the double jump Powerup in effect
 3. MAJOR: strong jump power, performed by landing on a LandTile with a jump pad
- ii. Obtainables (may be picked up by the Player)
 1. Double jump: allows the player to jump once midair, appearing in-game as a flashing green circle. Lasts for 30 seconds
 2. Invincibility: prevents the player from being killed in any way whatsoever, appearing in-game as a flashing red circle. Lasts for 30 seconds.
 3. Coin: grants +500 score, appearing as a large, "spinning" gold circle.
- iii. Collisions:
 1. Collision with the top surface of a LandTile sets the Player's vertical velocity to 0.
 2. Collision with a wall, an obstacle, or the undersurface of a midair LandTile will kill the Player and end the game.
- c. Score - for every frame that passes, the score will be incremented by 1 as long as the game is not paused and the game is not over (i.e. Player has not died).

B. Aesthetics

- a. Animations:
 - i. Ground animation to help simulate horizontal movement.
 - ii. A rolling animation is performed while the Player is midair, and corrects the Player's orientation appropriately upon landing.
 - iii. Varying width of Coins to simulate spinning.
 - iv. Powerups flash white at certain intervals.
 - v. The Player's avatar flashes red while the invincibility Powerup is in effect.
 - vi. A green trail follows the Player's movements while the double jump Powerup is in effect.
- b. Sound effects for:
 - i. Pressing buttons in the menus
 - ii. Obtaining coins and Powerups
 - iii. Player death
 - iv. Beginning a new run
- c. Soundtrack for main menu and a shuffling playlist (of 5 tracks) while in a run

C. Others

- a. Keybinds:
 - i. Q - pause game (while within a run)
 - ii. A - restart game at current gamemode (while within a run)
 - iii. Space - Player jump
- b. Menus
 - i. Main menu, which provides options (via Buttons) for quitting the game and choosing between one of four gamemodes:
 - 1. Classic: start the game off at easy difficulty, and the game becomes more difficult the longer the Player survives (regardless of score)
 - 2. Easy: constant easy difficulty
 - 3. Medium: constant medium difficulty
 - 4. Difficult: constant difficult difficulty
 - ii. Pause menu - allows the player to restart the game within the same gamemode, return to the main menu, or resume the current run
 - iii. Death menu - allows the player to restart the game within the same gamemode, return to the main menu, or quit the game

Brief How-to-play:

1. Select the difficulty from the main menu (classic for default difficulty progression, easy, medium, or difficult for constant respective difficulty)
2. Press space to jump — you do not have to press space to be boosted by jump pads (which appear as cyan slabs on the floor), but you do have to press space to be boosted by JumpOrbs (which appear as floating cyan circles). Time jumps to dodge obstacles (walls, spikes, and ceilings) and collect powerups and coins.
3. Red Powerups give 30-second invincibility — invincibility makes spikes and ceilings bouncy and wall collisions teleport the player's avatar into the air, far above any obstacles), and green Powerups give a 30-second double jump ability, allowing you to jump a second time midair. This second jump does not replenish until your avatar hits the ground.

UML Diagram (on the next page):

